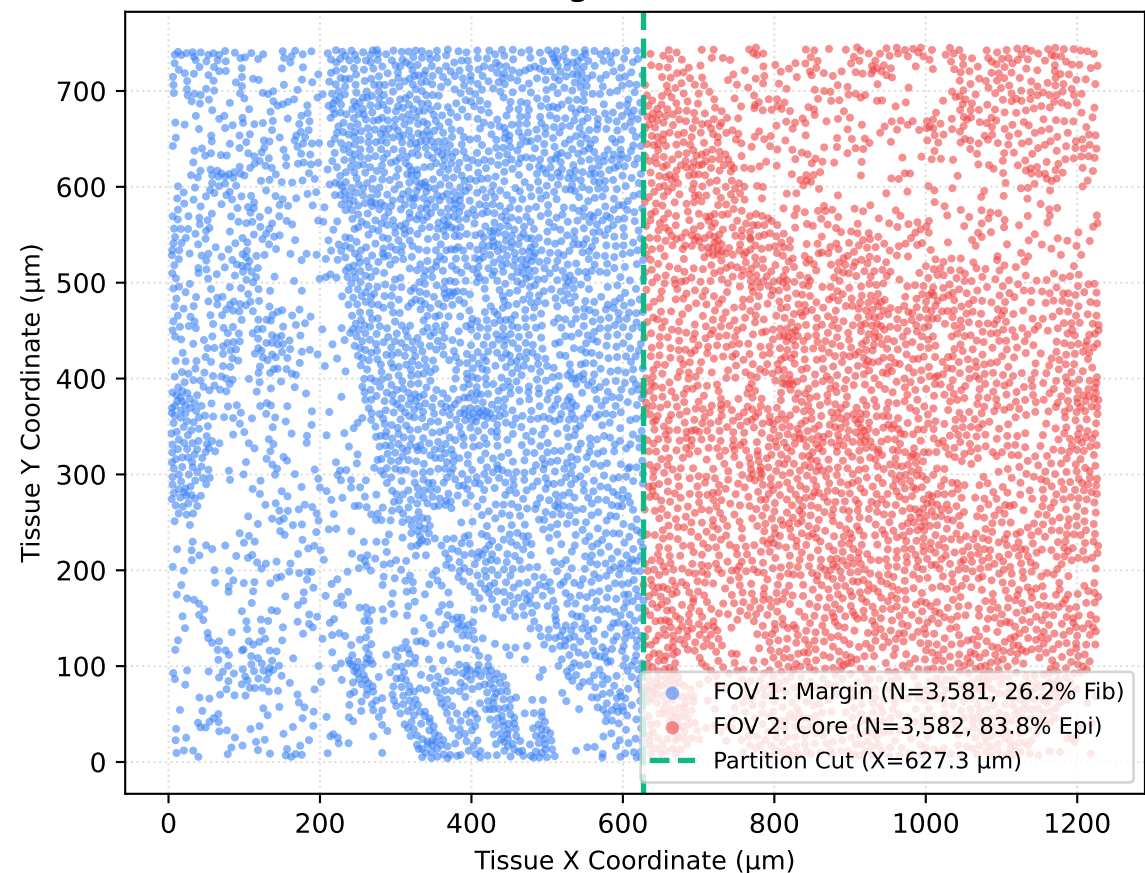
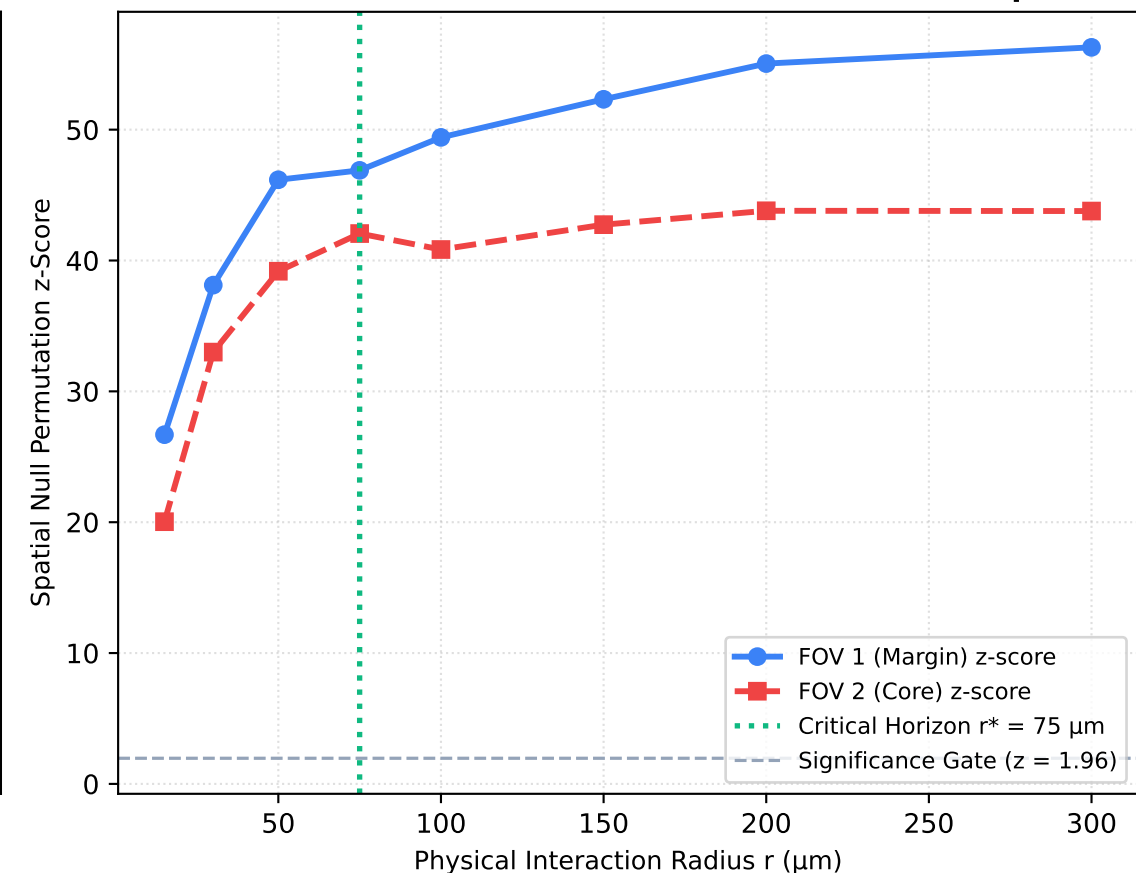


Cross-FOV Microenvironmental Replication: Critical Horizon & Single-Cell Extinction Stability

A: Cross-Region Spatial Partition
Stroma-Rich Margin vs Dense Tumor Core



B: Signaling Horizon Replication
CXCL12 Permutation z-Score vs Radius (15 to 300 μm)



C: Single-Cell Action Half-Distance Replication
Consistency Across Divergent Tumor Stroma Densities

